2025 International Conference on Advances in Modern Age Technologies for Health and Engineering Science (AMATHE) | 979-8-3315-0103-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/AMATHE65477.2025.11081248

Adaptive Concept Drift Detection Using Bayesian Neural Networks and Wasserstein Distance

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***Abstract*—Concept drift in streaming data poses a significant challenge to the stability and performance of deep learning models. This study explores an uncertainty-based drift detection approach leveraging Wasserstein Distance in a Bayesian Neural Network (BNN) framework. By continuously monitoring uncertainty variations across batches, the proposed method effectively identifies changes in data distribution. Experimental results demonstrate that lower drift thresholds lead to increased drift detection sensitivity, while higher thresholds result in more conservative drift identification. The findings validate Wasserstein Distance as a robust metric for real-time concept drift detection, providing a balance between model adaptability and stability. The study further highlights the necessity of adaptive learning strategies to handle evolving data streams.**

***Index Terms*—Concept Drift, Bayesian Neural Networks, Wasserstein Distance, Uncertainty Estimation, Adaptive Learning, Drift Detection, Streaming Data, Deep Learning**

I. Introduction

In real-world applications, machine learning models often operate in dynamic environments where data distributions evolve over time, leading to a phenomenon known as concept drift [1]. Concept drift occurs when the statistical properties of the target variable change, causing the predictive model to become outdated and lose accuracy. This problem is particularly prevalent in domains such as financial forecasting, industrial IoT monitoring, healthcare diagnostics, and fraud detection, where real-time decision-making is crucial.

Traditional machine learning models assume that training and test data share the same distribution. However, in non-stationary settings, this assumption fails, making drift detection and adaptation essential. Bayesian Neural Networks (BNNs) have gained attention as they provide uncertainty estimation, which can be leveraged for detecting drift in streaming data environments. In this study, we employ Wasserstein Distance, a statistical measure of distributional differences, to quantify uncertainty variations over time and detect drift effectively.

This work focuses on analyzing the impact of Wasserstein Distance-based uncertainty estimation in concept drift detection. The proposed method continuously monitors model uncertainty across batches and flags instances where Wasserstein Distance exceeds a predefined threshold, signaling potential drift. To evaluate the approach, we conducted experiments with different drift thresholds and analyzed their impact on drift detection sensitivity. The results demonstrate that lower thresholds capture minor drifts, whereas higher thresholds allow only significant drifts to be detected, emphasizing the trade-off between sensitivity and robustness. Figure 1 illustrate the phenomena of Concept drift, where the model learning becomes obsolete upon encountering the change in relationship between features and class labels due to the non-stationary nature of the learning environment. The blue colored learning indicate the historical learning which may be treated as Concept-1, after the drift point, the green colored learning indicates the emergence of Concept-2, The Drift happened from Concept-1 to Concept-2 indicating a drift.

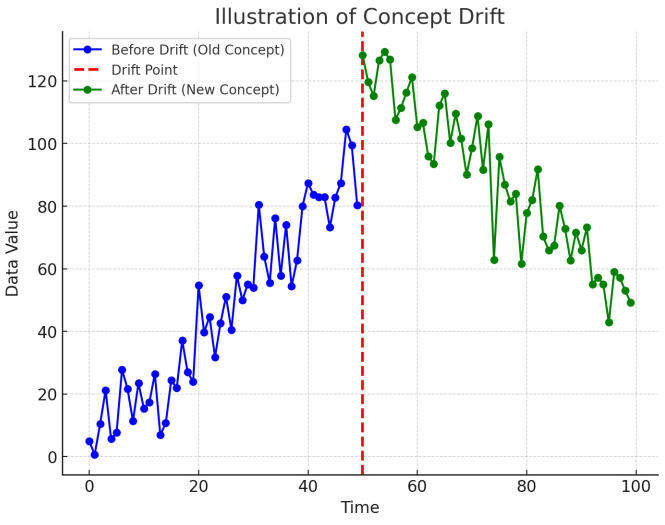


Fig. 1. Concept Drift illustration

The research question addressed in the presented work are,

1) *How can Bayesian Neural Networks (BNNs) be used to quantify and embody uncertainty for effective concept drift detection in online stream setting?*

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2) *Can Wasserstein Distance be used as an effective drift detection mechanism in Drift, Uncertainty aware Deep learning architectures for the effective detection, mitigation and adaptation of concept drift detection?*

The work presents Bayesian Neural Networks (BNNs) with Wasserstein Distance for uncertainty-driven learning in a nonstationary learning setting (Online Learning). BNNs uses probabilistic weight distributions instead of fixed weight parameters used in traditional deep learning models enabling uncertainty estimation be it epistemic and aleatoric. Wasserstein Distance is a efficient metric to detect distributional changes to enable dynamic drift detection without requiring explicit drift labels. Furthermore this work provides Wasserstein-based loss regularization mechanism that penalizes deviations introduced drift thus stabilizing the learning process even in the non-stationary environment where the drift is quite frequent.

The objectives of our work are,

1) To implement a Bayesian Neural Network (BNN) with Monte Carlo dropout for uncertainty estimation and prediction
2) Incorporate Wasserstein Distance between uncertainty distributions of current and historical batches and use it as a measure for Concept drift detection
3) Implement an adaptive learning strategy to not forget historical learning and retrain the model efficiently upon detecting drift

The rest of the sections are organized as follows: Section II examines the most relevant literature in the context of uncertainty-based concept drift detection. Section III gives a detailed discussion on the dataset used for the experimentation, whereas Section IV gives the learner architecture and provides background for Wasserstein distance. Section V presents the proposed methodology, whereas Section VI and Section VII provide a thorough examination of the results obtained and present the conclusion of the proposed work, respectively.

II. Background

The literature survey was conducted using a systematic review approach to identify relevant research on concept drift detection using Bayesian neural networks and Wasserstein distance. The methodology involved querying multiple academic databases, including IEEE Xplore, Springer, ScienceDirect, ACM Digital Library, and arXiv, using specific keywords such as "Concept Drift Detection," "Bayesian Neural Networks," "Wasserstein Distance," "Uncertainty Estimation in Concept Drift," and "Adaptive Learning for Concept Drift". The selection criteria prioritized papers published in peer-reviewed journals, top-tier conferences, and preprint archives from 2010 to 2024. The papers were then filtered based on relevance to the research domain, with a focus on methodological contributions, datasets used, and evaluation metrics (accuracy, precision, and F1-score). The final selection consisted of ten highly relevant papers, spanning applications in time series, IoT, industrial processes, and machine learning-based classification tasks. The contributions of each study were extracted, and a comparative analysis was conducted to summarize their approaches, findings, and performance metrics.

The research in [2] presents an approach called Dynamic Logistic Regressor (DLR) involves training models on blocks of consecutive samples and using their predictions to update beliefs about concept drift points. To ensure the consistency of the model, the work removes the drift points with lower belief or probability. The authors in [3] introduced a novel approach called WATCH. WATCH, uses the Wasserstein distance to model the initial data distribution and track its periodic changes as new data points are processed. The researchers have used highly complex time series data to examine its performance. WATCH outperformed state-of-the-art change point detection methods in terms of accuracy and reliability.

Researcher in [4] introduces an improved Wasserstein distance-based transfer learning method called WDA for machine fault diagnosis. By leveraging Wasserstein distance with cosine similarity, WDA effectively transfers features between labeled and unlabeled signals from different equipment. Experimental results across multiple transfer scenarios show that WDA achieves high accuracy (93.72% for bearing faults and 84.84% for ball screw faults), outperforming state-of-the-art fault diagnosis methods.

The research at [5] presented an improved anomaly detection on sensor data by testing regression-based methods, Bayesian autoencoders, and statistical tests. The author presented the combined model with Kullback-Leibler divergence, effectively detecting gradual long-term drifts. To stabilize the output , adaptive windowing technique is suggested.

The authors in [6] introduced Partial Wasserstein Distance and Maximum Mean Discrepancy (MMD) Distance to combine outlier detection and drift detection. Partial Wasserstein Distance captures distributional shifts while ignoring minor changes, making it robust against noise in the data. MMD gives differences between probability distributions in high-dimensional spaces, and used for detecting gradual drifts.

The work in [7] presents experimental analysis of various dissimilarity metrics to evaluate their effectiveness in detecting sudden concept drift. Distance measures such as Wasserstein Distance, Kullback-Leibler Divergence, and Maximum Mean Discrepancy (MMD) and assessed their sensitivity to abrupt changes in data distributions. Strength and Weaknesses are presented.

The research in [8] examines Domino Drift Effect Approach for estimating the probability of feature drift in high-dimensional data. By using probability estimation techniques, it provides a structured approach for detecting and quantifying drift across complex datasets. Table I summarizes the findings of the literature survey.

III. Dataset Description

The dataset used in this experimentation is taken from OpenML open source repository [9] which is curated from the original source [10] which is a weather data compiled with Weka tool for Decision tree classifier which is rich in distributional shifts. The dataset comprises of 8 features

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TABLE I
SURVEY OF CONCEPT DRIFT DETECTION APPROACHES

Reference	Method	Key Contributions
[2]	Dynamic Logistic Regressor (DLR)	Trains models on consecutive data blocks and updates beliefs about concept drift points. Removes drift points with lower probability to maintain model consistency.
[3]	WATCH (Wasserstein-based concept drift detection)	Uses Wasserstein distance to model data distribution and track periodic changes. Evaluated on complex time series data, outperforming state-of-the-art change point detection methods.
[4]	Wasserstein Distance-based Transfer Learning (WDA)	Combines Wasserstein distance with cosine similarity for machine fault diagnosis. Achieves high accuracy in multiple transfer scenarios for bearing and ball screw faults.
[5]	Regression-based + Bayesian Autoencoders + KL Divergence	Proposes improved anomaly detection in sensor data. The best approach combines regression and Kullback-Leibler divergence with adaptive windowing for stable detection of gradual drifts.
[6]	Partial Wasserstein Distance + Maximum Mean Discrepancy (MMD)	Combines outlier detection and drift detection. Partial Wasserstein captures distribution shifts while ignoring noise, and MMD detects gradual drifts in high-dimensional spaces.
[7]	Experimental Analysis of Dissimilarity Metrics	Evaluates distance measures like Wasserstein Distance, KL Divergence, and MMD for sudden concept drift detection. Highlights strengths and weaknesses of each metric.
[8]	Domino Drift Effect Approach	Estimates feature drift probability in high-dimensional data using probability estimation techniques. Provides a structured approach for detecting and quantifying drift.

and binary class label, with a total of 8 columns and 18159 rows. The dataset overview is presented in the Table II. The features present in the dataset are, Temperature, DewPoint, SeaLevelPressure, Visibility, AverageWindSpeed, MaximumSustainedWindSpeed, Maximum and Minimum temperature with class label as binary boolean value. The features, its description and the type of value present are summarized in the Table III. Upon performing Exploratory data analysis, the statistical summary of each features are summarized in the Table IV. The dataset is imbalanced, with 12461 negative samples and 5698 positive samples. To balance the dataset, we have downsampled the data to 5698 positive and negative samples.

TABLE II
DATASET OVERVIEW

Statistic	Value
Number of Features	8
Number of Rows	18159
Number of Columns	9
Class Distribution (False)	12461
Class Distribution (True)	5698

The feature distribution is shown in figure 2. The feature correlation matrix which gives insights on the relationship that exist between various features are shown in figure 3.

The figure 4 represents the p-value obtained from Kolmogorov-Smirnov (KS) test and T-test across 50 batches of 100 samples each. The dashed line in black indicates the drift threshold which is set to 0.05. If the p-value is above 0.05, that indicates the distributions are similar, where as anything below the threshold indicates the potential of distribution change inferring Concept Drift. Referring to the Figure 4, we can see that most of the batches have p-value less than 0.05, except

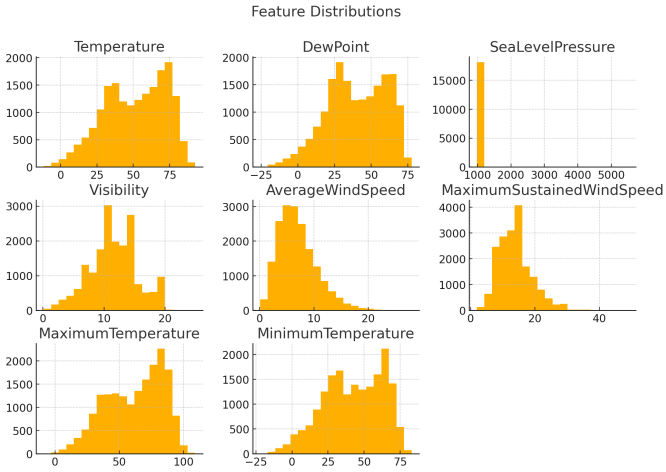


Fig. 2. Weather dataset-Feature Distribution

few batches that is batch number 9, 31, 38 making the dataset a suitable real-world dataset for the experimentation.

IV. BAYESIAN NEURAL NETWORK AND UNCERTAINTY

Bayesian Neural Networks (BNNs) introduced by [11] extend traditional neural networks by incorporating probability distributions over weights, allowing for uncertainty estimation in classification tasks. Instead of deterministic weight parameters, BNNs maintain a posterior distribution over the weights given the data.

A. Bayesian Inference in Neural Networks

Given a dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ consisting of input-output pairs, a standard neural network estimates the likelihood $p(y|x, w)$, where w represents the weights of the network. In

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TABLE III
FEATURE DESCRIPTIONS FOR THE WEATHER DATASET

Feature Name	Type	Description
Temperature	Float	Atmospheric temperature (°C)
DewPoint	Float	Temperature at which air is saturated (°C)
SeaLevelPressure	Float	Pressure at sea level (hPa)
Visibility	Float	Distance visible in atmosphere (km)
AverageWindSpeed	Float	Average wind speed (km/h)
MaximumSustainedWindSpeed	Float	Maximum sustained wind speed (km/h)
MaximumTemperature	Float	Maximum recorded temperature (°C)
MinimumTemperature	Float	Minimum recorded temperature (°C)
Class	Boolean	Binary label indicating weather condition (True/False)

TABLE IV
STATISTICAL SUMMARY OF WEATHER DATASET

Feature	Count	Mean	Std	Min	25%	50%	75%	Max
Temperature	18,159	51.01	21.15	-11.70	34.60	53.00	69.60	92.50
DewPoint	18,159	39.74	20.08	-25.70	25.30	39.70	57.45	77.60
SeaLevelPressure	18,159	1016.76	47.70	980.20	1011.40	1015.90	1020.80	5503.80
Visibility	18,159	11.66	3.94	0.10	9.30	11.40	14.50	24.90
AverageWindSpeed	18,159	7.02	3.66	0.10	4.40	6.40	9.00	28.00
MaximumSustainedWindSpeed	18,159	14.02	5.32	1.90	10.10	13.00	17.10	49.00
MaximumTemperature	18,159	62.03	22.18	-8.90	44.10	64.90	81.00	109.00
MinimumTemperature	18,159	42.47	20.44	-22.00	28.00	43.00	60.10	82.90

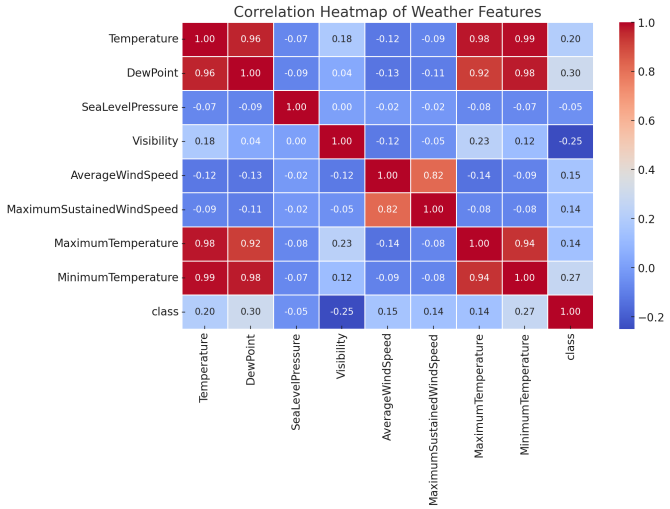


Fig. 3. Weather dataset-Feature Correlation Matrix

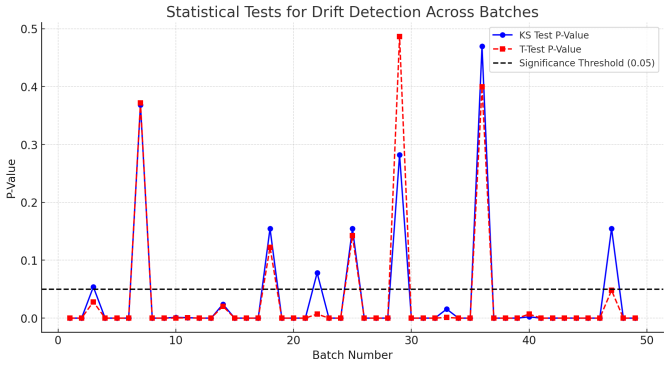


Fig. 4. Statistical Test Results for Concept Drift confirmation

a Bayesian framework, we define a posterior distribution over weights:

$$p(w|\mathcal{D}) = \frac{p(\mathcal{D}|w)p(w)}{p(\mathcal{D})} \tag{1}$$

where: - $p(w|\mathcal{D})$ is the posterior distribution over weights, - $p(\mathcal{D}|w)$ is the likelihood function, - $p(w)$ is the prior distribution over weights, - $p(\mathcal{D})$ is the marginal likelihood (evidence).

Since computing the posterior distribution exactly is intractable for deep networks, approximation techniques such as Monte Carlo Dropout [12], Variational Inference, or Markov Chain Monte Carlo (MCMC) are used.

B. Variational Inference for Approximate Posterior Estimation

Variational Inference (VI) approximates the posterior $p(w|\mathcal{D})$ with a simpler distribution $q(w|\theta)$ by minimizing the Kullback-Leibler (KL) divergence:

$$D_{KL}(q(w|\theta)||p(w|\mathcal{D})) = \mathbb{E}_{q(w|\theta)}\left[\log \frac{q(w|\theta)}{p(w|\mathcal{D})}\right] \tag{2}$$

Since the evidence $p(\mathcal{D})$ is intractable, we maximize the Evidence Lower Bound (ELBO):

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$$\mathcal{L}(\theta) = \mathbb{E}_{q(w|\theta)}[\log p(\mathcal{D}|w)] - D_{KL}(q(w|\theta)||p(w)) \quad (3)$$

where: - The first term represents the expected log-likelihood of the data under the approximate posterior, - The second term regularizes the approximate posterior to be close to the prior.

C. Uncertainty Estimation with Monte Carlo Sampling

To make uncertainty-aware predictions, we approximate the predictive distribution by drawing T samples from the posterior:

$$p(y|x, \mathcal{D}) = \mathbb{E}_{p(w|\mathcal{D})}[p(y|x, w)] \approx \frac{1}{T}\sum_{t=1}^{T} p(y|x, w_t) \quad (4)$$

where $w_t \sim q(w|\theta)$ are sampled from the approximate posterior.

The uncertainty [13] in the prediction can be decomposed into: - Aleatoric Uncertainty (data uncertainty): due to inherent noise in the data. - Epistemic Uncertainty (model uncertainty): due to lack of knowledge in model parameters.

The total predictive variance is:

$$\text{Var}(y|x, \mathcal{D}) = \underbrace{\mathbb{E}_{p(w|\mathcal{D})}[\text{Var}(y|x, w)]}_{\text{Aleatoric Uncertainty}} + \underbrace{\text{Var}_{p(w|\mathcal{D})}[\mathbb{E}(y|x, w)]}_{\text{Epistemic Uncertainty}} \quad (5)$$

D. Concept Drift Detection Using Wasserstein Distance

For online learning settings, concept drift detection can be formulated using the Wasserstein Distance between uncertainty distributions over a reference window and incoming batches.

Given two probability distributions, P (reference uncertainty) and Q (current batch uncertainty), the Wasserstein Distance is:

$$W(P, Q) = \inf_{\gamma \in \Pi(P,Q)} \mathbb{E}_{(p,q)\sim\gamma}[\|p - q\|] \quad (6)$$

where $\Pi(P, Q)$ is the set of joint distributions with marginals P and Q.

A drift is flagged when:

$$W(P, Q) > \delta \quad (7)$$

where δ is a predefined threshold. This enables adaptive retraining of the model in response to evolving data distributions.

E. Training Objective with Wasserstein-Based Adaptation

To enhance the robustness of online learning, we modify the BNN loss function to incorporate drift adaptation using Wasserstein Distance:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BNN}} + \lambda W(P, Q) \quad (8)$$

where:

- $\mathcal{L}_{\text{BNN}}$ is the standard BNN loss (e.g., Binary Cross-Entropy).
- $W(P, Q)$ penalizes distributional shifts
- λ is a weighting factor

By integrating Bayesian inference with Wasserstein-based drift detection, we achieve adaptive concept drift handling in uncertainty-aware classification.

V. Proposed Methodology

The figure 5 shows the proposed methodology of our presented work. The overall process starts by preprocessing the real-world drift dataset(Weather), which includes, converting the features into tensors and normalizing the values to make it ready for processing by Deep learning architectures. The data is reorganized into small stream of batches, to make it suitable for online learning setting. The learner used for uncertainty based, drift aware based concept drift detection is BNN, which incorporates Monte-Carlo dropout for each batch to estimate model uncertainty which is stored in the reference window. The Wasserstein Distance is used as drift detection measure is then computed between the uncertainty distribution of the current batch and the reference window. If the Wasserstein Distance is greater than drift threshold, then it indicates a potential concept drift, thus triggering the adaptation mechanism of retraining the model on the newest batch and updating the BNN in turn before taking in next batch of data. If the value is below threshold that indicates that model did not experience uncertainty which will allow the model to take in the next batch for processing. The approach is adaptive in nature, capable of learning evolving data distribution in the non-stationary learning environment.

The algorithm 1 summarizes the proposed methodology discussed above in a crisp manner.

VI. Results and Discussion

The experimentation was conducted in Acer Nitro-5 enabled with Ryzen 7 CPU and Nvidia Geforce RTX GPU 350. The result obtained for the drift threshold=0.000065 is shown in figure 6. In the figure, the blue line indicate the BNN's uncertainty value over streaming batches over time, whereas the dashed line representing the drift threshold. Any batches which has its Wasserstein Distance greater than drift threshold indicates that the concept drift has happened in those batches and drift adaptation was triggered to mitigate the drift. The identified drift points are marked in red cross mark, evidently for 0.000065 drift threshold, 12 drift points were identified. From the graph its evident that concept drift occurred are non-uniform in nature, and the data distribution is changing

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Algorithm 1 Online Concept Drift Detection using BNN and Wasserstein Distance

Require: Dataset $\mathcal{D} = \{X, y\}$, batch size B, reference window size W, drift threshold δ

Normalize features using StandardScaler
Convert X and y to PyTorch tensors
Initialize Bayesian Neural Network (BNN)
Define loss function (Binary Cross-Entropy) and optimizer (Adam)
Initialize reference window $U \leftarrow \emptyset$ and drift points $D \leftarrow \emptyset$
for each batch b in dataset $\mathcal{D}$ **do**
    Extract batch X_b, y_b
    Compute uncertainty using Monte Carlo Sampling
    if $|U| < W$ **then**
        Store batch uncertainty in reference window U
    else
        Compute Wasserstein Distance W_d between current batch and reference window
        if $W_d > \delta$ **then**
            Drift detected, add batch index to D
            Retrain BNN using batch X_b, y_b
            Update model parameters using Gradient Descent
        end if
        Update reference window U with latest uncertainties
    end if
end for
return updated Bayesian Neural Network

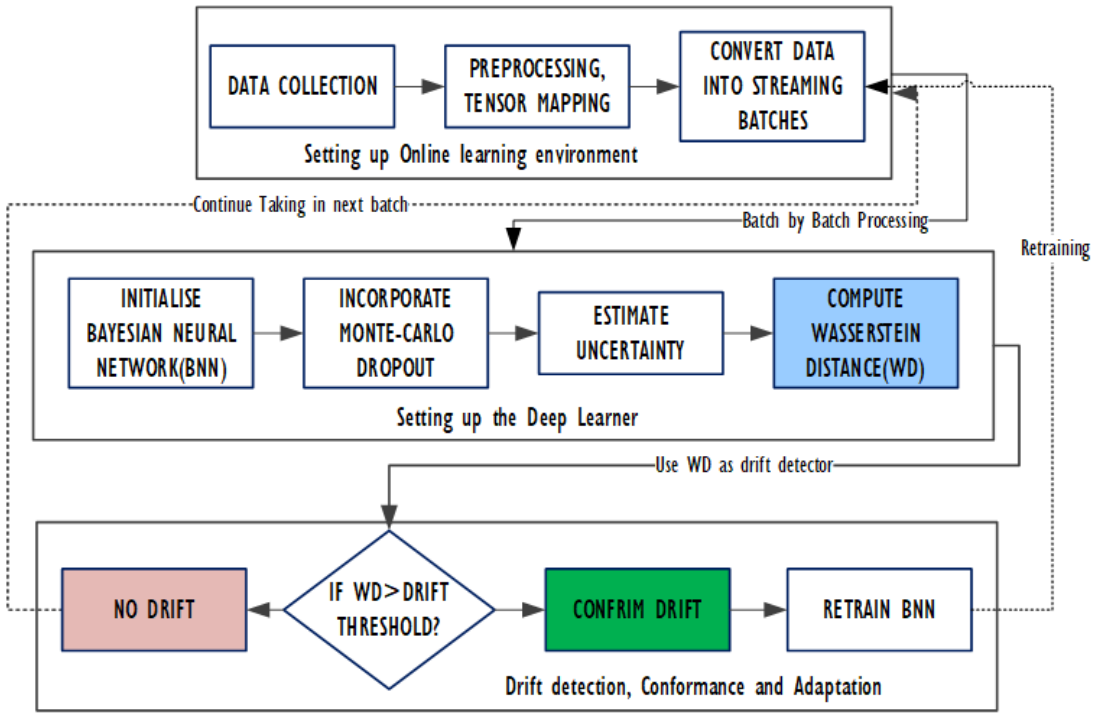


Fig. 5. Proposed Methodology

at regular intervals due to environmental factors. The sharp spikes observed indicates that the model experienced high uncertainty from previous data patterns. From this analysis, Wasserstein Distance proved to be an effective measure for concept drift detection in probabilistic deep learning models like BNNs which use MC-dropout and use uncertainty estimation. The model gets into training every-time the Wasserstein Distance crosses the drift threshold, which is indicated by dip in the uncertainty after every spike in the figure 6.

Varying the drift threshold did show lot more improvements,

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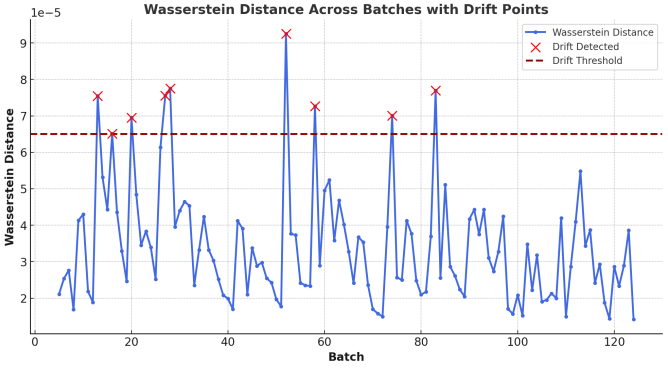


Fig. 6. Wasserstein Distance vs Batch Number graph

as model try to stabilize and not to reach much on small distributional change. In simple words, the smaller threshold had higher detection of drift points, where as higher thresholds lessened the number of drift points flagged by the model. The table V for 500, 50 being the window and batch size respectively. Figure 7 summarizes the findings by varying the drift threshold and identified number of drift points for reference.

TABLE V
DRIFT DETECTION RESULTS WITH DIFFERENT DRIFT THRESHOLDS

Window Size	Batch Size	Drift Threshold	#Drift Points
500	50	0.000075	8
500	50	0.000065	13
500	50	0.000060	89
500	50	0.000055	48
500	50	0.000050	264

Fixing the batch size as constant and drift threshold and varying Windows size, evidently decreased the number of drift points detected as shown in table VI. Similarly, fixing the Window Size and varying the drift threshold with doubling batch size actually increased the drift points making the model sensitive to smallest change in the distribution under study. Figure 7 and 8 illustrates the findings in the form of plots for better visualization.

TABLE VI
DRIFT DETECTION RESULTS WITH VARYING WINDOW SIZES

Window Size	Batch Size	Drift Threshold	#Drift Points
100	100	0.000065	26
200	100	0.000065	12
300	100	0.000065	13
400	100	0.000065	8
500	100	0.000065	12

VII. CONCLUSION AND FUTURE SCOPE

This study demonstrated an effective approach for concept drift detection using Wasserstein Distance and Bayesian Neural Networks (BNNs). The results highlight that uncertainty quantification through Wasserstein Distance provides a robust

TABLE VII
DRIFT DETECTION RESULTS WITH VARYING DRIFT THRESHOLDS

Window Size	Batch Size	Drift Threshold	#Drift Points
500	100	0.000045	78
500	100	0.000050	98
500	100	0.000055	20
500	100	0.000060	53
500	100	0.000065	12

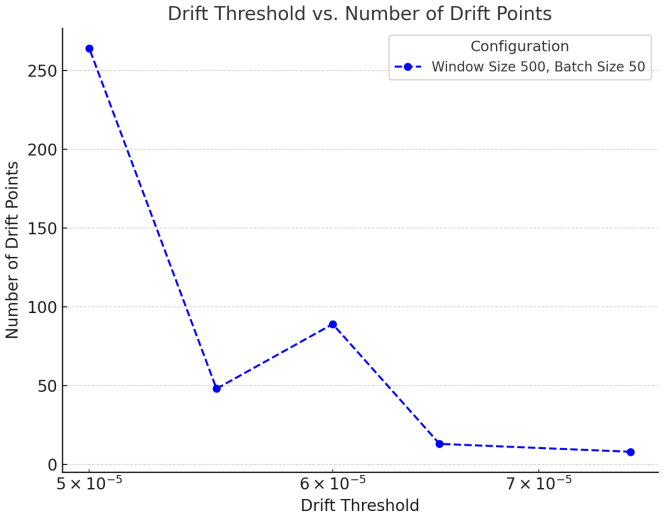


Fig. 7. Drift threshold vs Number of Drift points

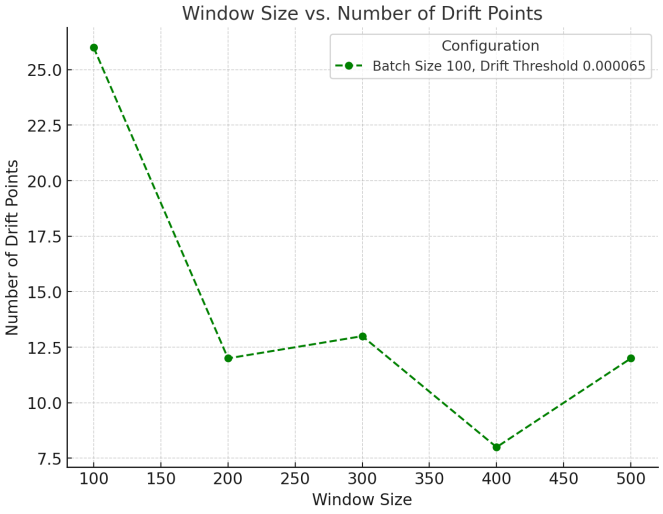


Fig. 8. Window Size vs. Number of Drift Points

mechanism for detecting drift in streaming data environments. The experiment confirmed that as the drift threshold increases, the number of detected drift points decreases, indicating a trade-off between sensitivity and robustness in detecting drifts. The visualization of Wasserstein Distance across batches effectively localized drift occurrences, validating the effectiveness of the proposed method in real-time applications. This experiment also emphasizes the importance of adaptive learning

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mechanisms in deep learning models, ensuring they remain resilient to evolving data distributions.

While the proposed method successfully detects and localizes concept drift, several enhancements and future research directions can be explored,

- Instead of a fixed threshold, implementing dynamic thresholding based on real-time statistics can improve drift detection accuracy
- Extending the model to differentiate between minor and major drift events can allow for selective adaptation strategies
- Exploring continuous model retraining strategies such as meta-learning and reinforcement learning-based adaptation can enhance long-term model stability
- Enhancing interpretability by integrating explainable AI (XAI) techniques can provide deeper insights into why drift occurs and how it affects model performance.

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